



SMARTSTAGE

SMART PATH SPLITTING & SUSTAINABLE TRAVEL

The Starting Point

"By 2030, devise and implement policies to promote sustainable tourism that creates jobs and promotes local culture and products."

- UN SDG Target 8.9



Our Shared Passion

As a team of developers and outdoor sports enthusiasts, we wanted to contribute directly to this goal.



Focus on Active Travel

We chose to promote multi-day trekking and cycling journeys, which have **minimal environmental impact**.



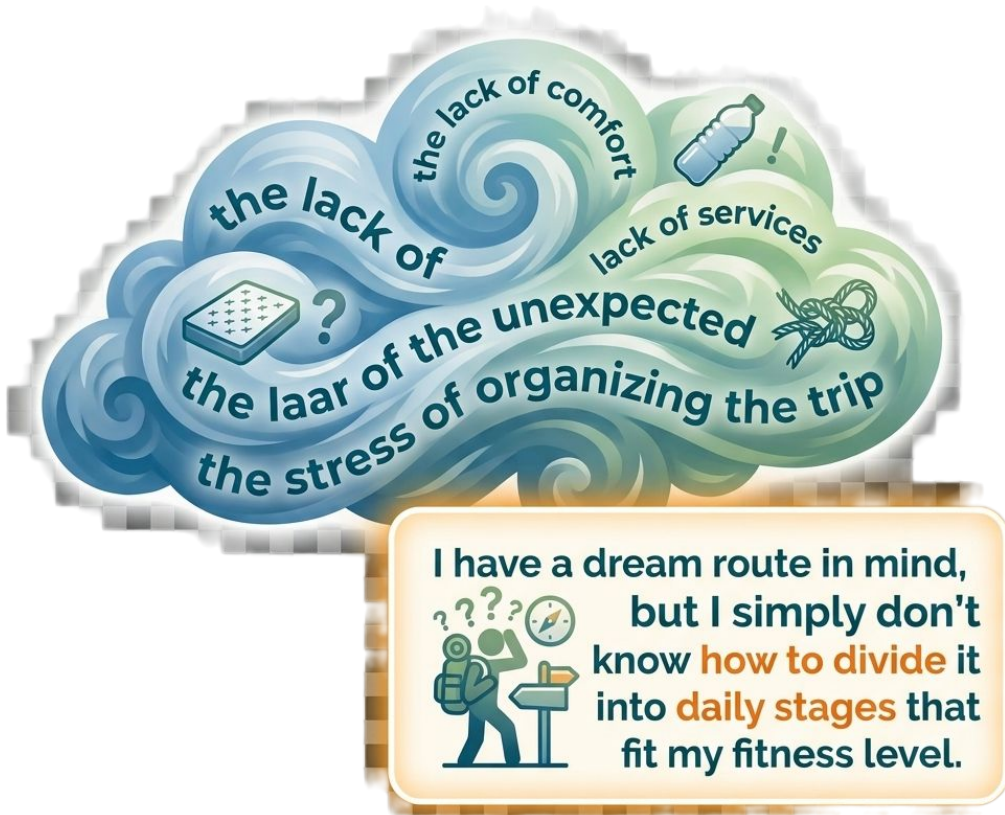
The Goal

Encourage more people to discover the beauty of slow, sustainable travel with **multi-day walking and cycling trips !**



The Problem

"Why don't more people do multi-day walking and cycling trips?"



The Solution:

SMARTSTAGE YOUR TAILOR-MADE ASSISTANT

ANALYZING FITNESS TRACKER



analysis from past workouts calculating performance level, defining effort

DREAM ROUTE UPLOAD



uploading .gpx file of the trip you want to do

PERSONALIZED STAGE SPLITTING



divides your long routes into tailor-made daily stages

SMART PATH SPLITTING & SUSTAINABLE TRAVEL

Lacks technical planning skills. The fear of getting stranded or physically exhausted prevents her from actually starting the journey.

DEFINING THE TARGET

HUMAN-CENTERED DESIGN: USER PERSONA



Marco, 32

The Aspiring Adventurer

“

I dream of completing a multi-day trek, but I'm terrified of overexerting myself because I have no idea how to safely plan the daily stages.



Profile

An office worker with low-to-medium athletic training. Tracks workouts with an Apple Watch but without knowledge about how to interpret its metrics



Motivations

Wants to safely complete his first **multi-day walking trip**. Highly motivated to experience nature and travel sustainably.



Pain Points

He is afraid he won't be able to **plan the stages of his route** in a way that's **appropriate for his fitness level**. He usually goes running on flat terrain, so he doesn't know how to handle the elevation



Needs

A tool that **plans the trip for him**, safely and easily. Needs reassurance that the proposed stages are perfectly tailored to his capabilities.

IDEATION PHASE

BRAINSTORMING FEATURES & FUNCTIONALITIES

Interactive Map
Integration

Reward points for rest
stops

Display city names on
map

Personal User Database
(Cloud)

Balanced stage
splitting algorithm
(equal km/day)

Performance algorithm
including Heart Rate
(HR)

Performance algorithm
including daily non-
workout steps

End stages strictly
near accommodations

CORE APP FUNCTIONALITY

FEATURES & FUNCTIONALITIES

1 .Data-Driven Personalization:

Calculate the maximum distance in kilometers that the user can cover on a multi-day trip using 30-day server data



View personal User Metrics

Explore 30-day activity charts

2. Smart Route Splitting:

Upload a .gpx and auto-divide it based on your limits



Export .gpx for each stage

Manage & Edit Trips

Interactive Elevation Profiles

IMPLEMENTATION DECISIONS



1. Separated State

Used two distinct Providers (Training & Trips) for a cleaner, highly modular architecture.



2. Storing What Matters

Raw server data is instantly converted into User Metrics to optimize local storage.



3. Privacy & Flexibility

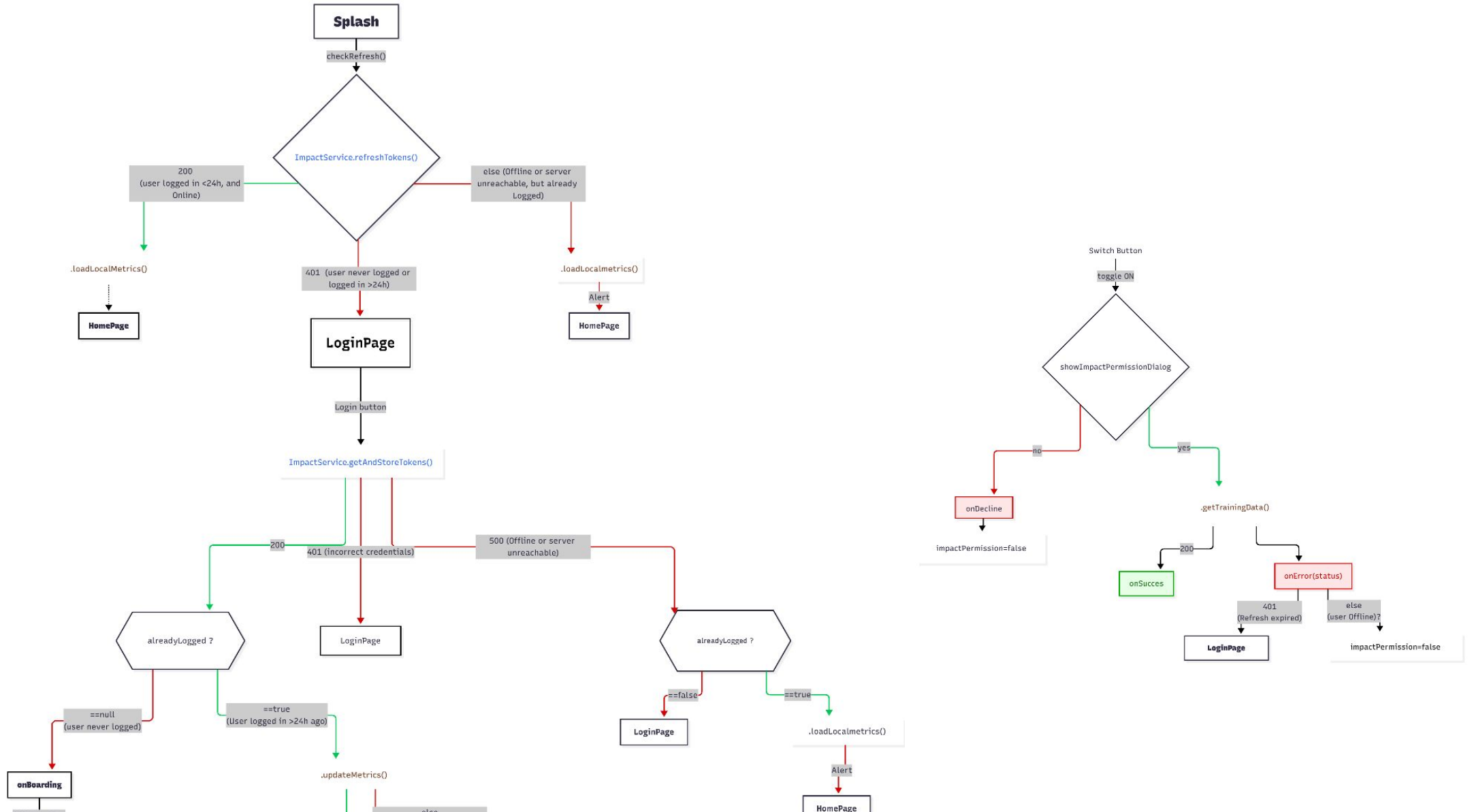
An inclusive app. It is fully functional even without fitness tracker consent via manual input.



4. Offline UX

Cached metrics and login states ensure a seamless user experience without internet.

Application Lifecycle & User Authentication Flow



Live Demonstration



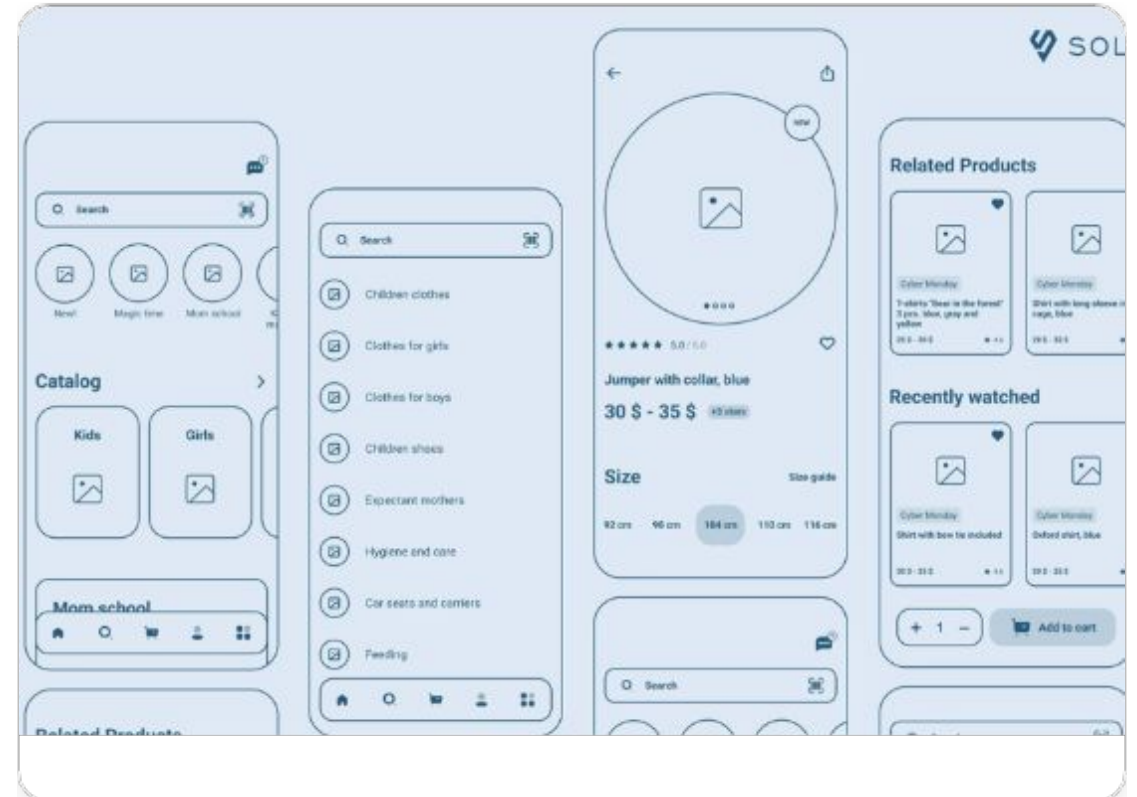
Bringing the concept to life

Iterative Design & User Testing

Learning from Version 1

We developed an initial prototype and tested it with real users matching your target persona. Using a co-design approach - observing them use the app and noting friction points.

- ✗ **Unclear Onboarding:** Users didn't know what "IMPACT Server" meant. We replaced all technical terminology for reduce cognitive load
- ✗ **No guidance on First Use:** Users landing on the Trips screen didn't know what to do. We added step-by-step instructions directly on the empty state screen.
- ✗ **Data Overload:** GPX upload lacked clear instructions on transport modes.



UI Evolution: V1 vs V3

Version 1 (Initial Prototype)

Complex terminology : The app used technical terms like “IMPACT” that most users didn’t understand.

No first-Use Guidance: Users landing on the Trips screen didn’t know what to do - no instructions, no empty state guidance.

Pool Visual Hierarchy: The Training screen emphasized secondary labels and charts, burying the most important information: the user’s daily km limits

Inconsistent Visual Language: Walk and bike represented with different colors and icons across screen, breaking visual consistency.

Version 3 (Final Release)

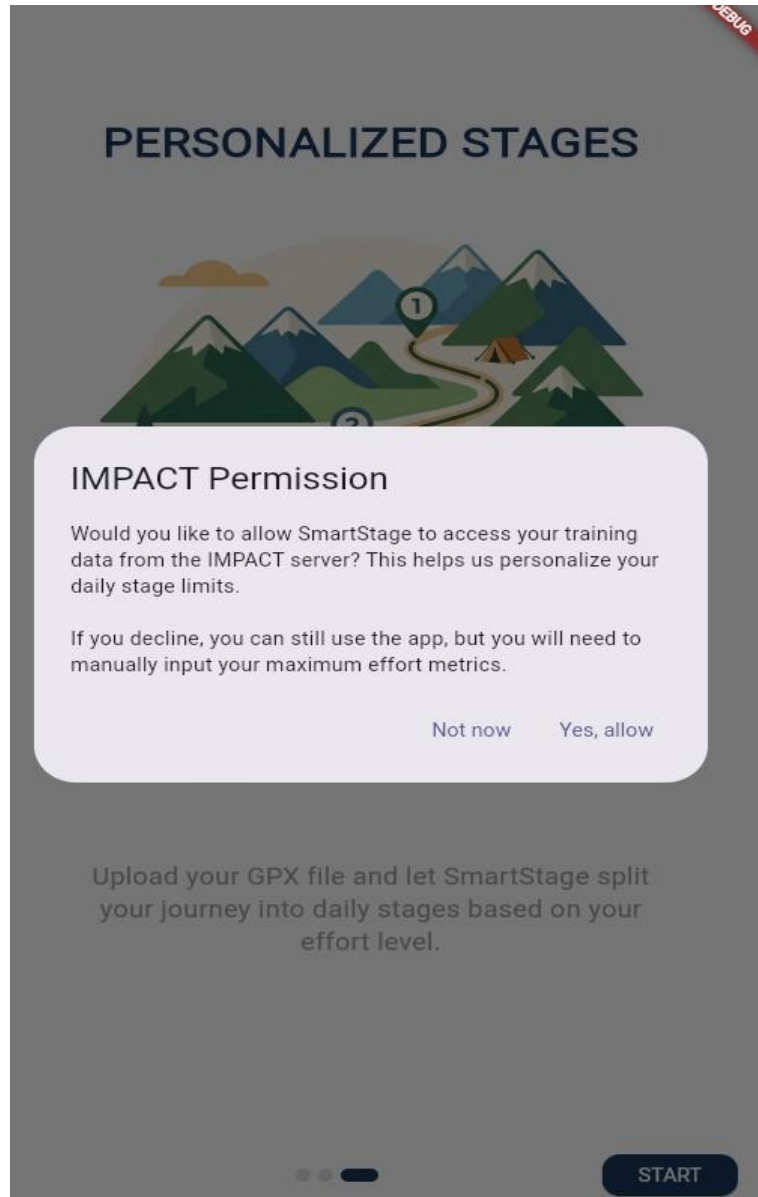
Fitness Tracker Language: Replaced server terminology with the familiar concept of a "fitness tracker" — zero technical knowledge required.

Guided Onboarding: Added a 3-step carousel explaining the app before asking any permission.

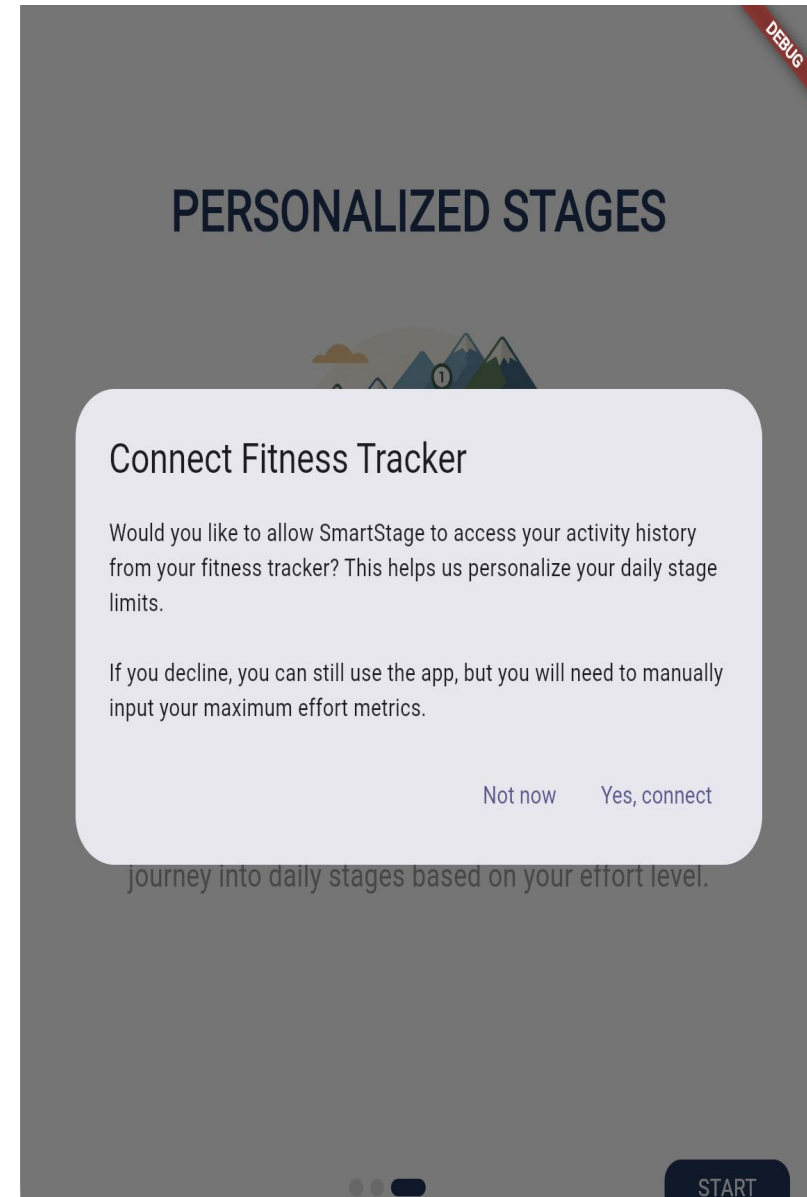
Clear Fallback: If tracker connection is denied, users are immediately routed to ManualEffortScreen to set limits manually.

Unified Color System: Orange always means walk, green always means bike — consistent across every screen.

version 1



version 3



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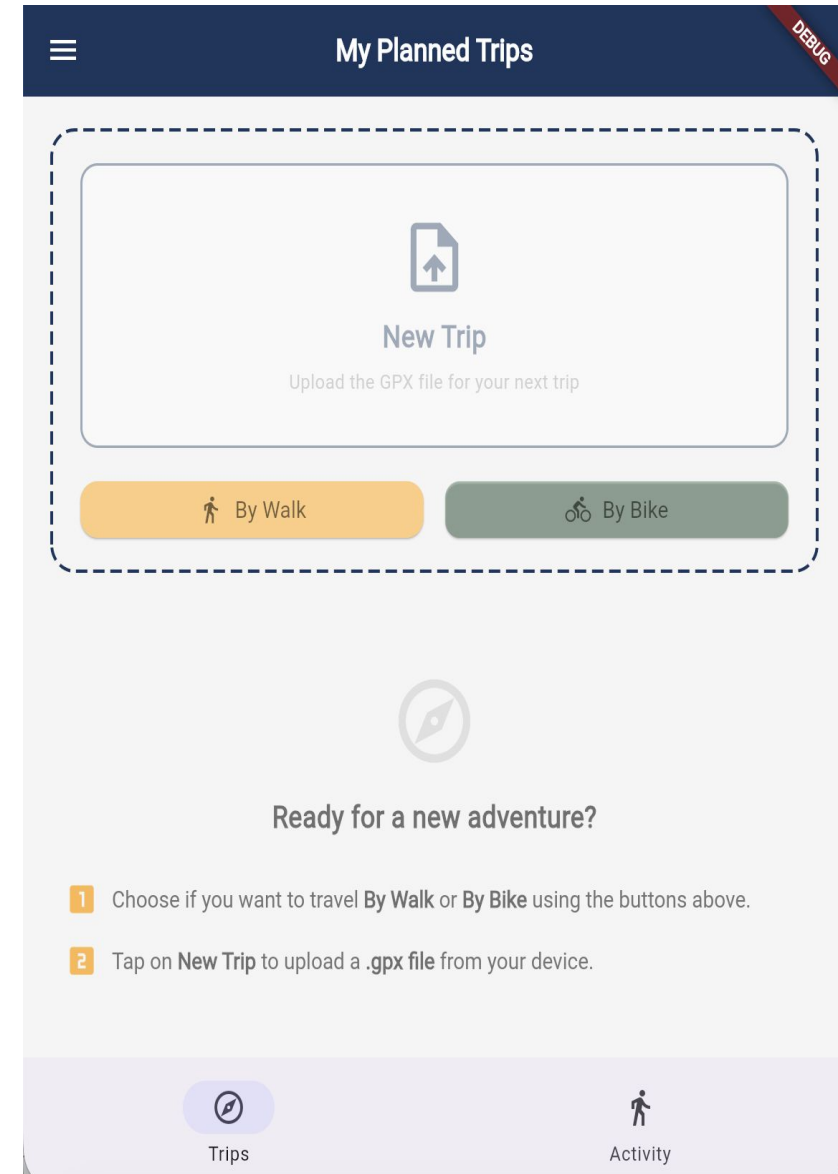
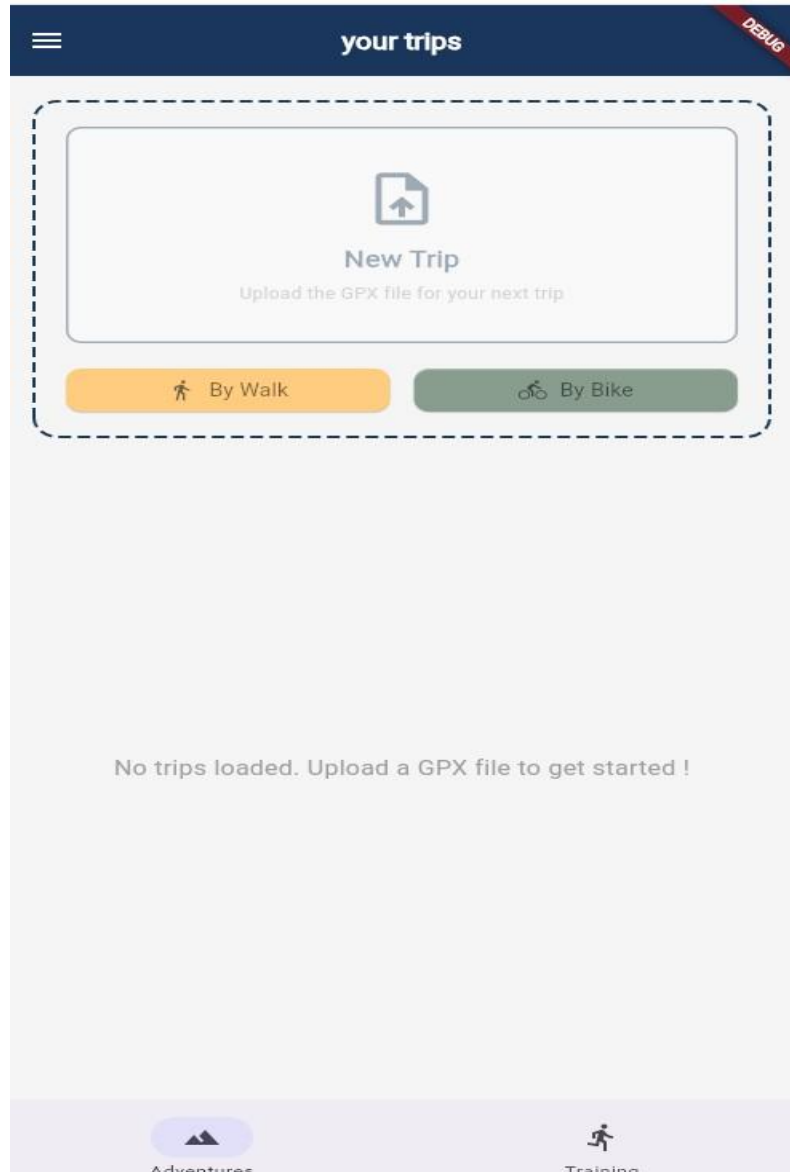
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Examples

version 3



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Limitations & Future Developments

Area	Current Limitation (V3)	Future Implementation
Performance Algorithm	Estimates effort using distance and elevation only. Ignores cardiovascular data (heart rate, pace, daily step count)	Integrated HR data from IMPACT to compute fatigue metrics (TRIMP , CTL, ATL) for a full physiological user profile.
Stage splitting	Cuts the route exactly at the mathematical effort threshold, with no awareness of whether the endpoint is near shelter, water or inhabited areas.	Smart splitting via Overpass/OpenStreetMap API to snap endpoints to POIs, plus load-balancing to distribute effort evenly across all stages.
Map & Persistence	No interactive 2D map view. Trip data is stored in memory only and lost when the app is closed	Integrate flutter_map for interactive GPX visualization with cut-point markers, and Firebase for cloud trip history.
Community	Trip plans are strictly local and cannot be shared. There is no social layer connecting users..	Add social feed where users can publish, review and discover beginner-friendly segmented routes shared by the community.

Questions?

Thank you for your attention.